# Concept Document/GDD

## Game Idea

- Genre : Real - Time Turned Based Strategy
- Dimension : Top - Down 2D.

### Core Mechanics

1. Grid Based Movement
    - The Playing field consists of a 25 × 17 grid, which the player can traverse through.
2. Chest System
    - Chest will have a area where defenses cannot be placed.
    - Once the player is on a chest the player will open a chest.
    - Once opened the player will be presented with a choice between what power ups to choose from with a set amount of coins to be returned.
3. Resource Management
    - Coins can be earned by destroying enemy defenses, attacking enemy kingdom, by the coin generator and by opening chests.
4. Store System
    - Defenses and anti - defense can be bought here, then eventually allowed to place.
5. Defenses
    - After each turn defense health will drop.
    - Their main objective is to destroy the enemy king's kingdom.
    - They will do damage each turn played.
    - They take a certain amount of space to be placed.

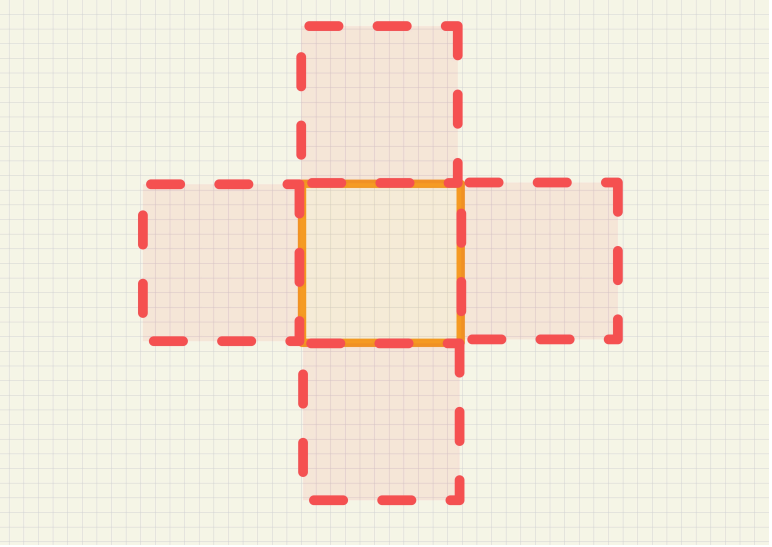
- The main threat they need to be aware of are anti-defenses who's main objective is to destroy enemy defenses.

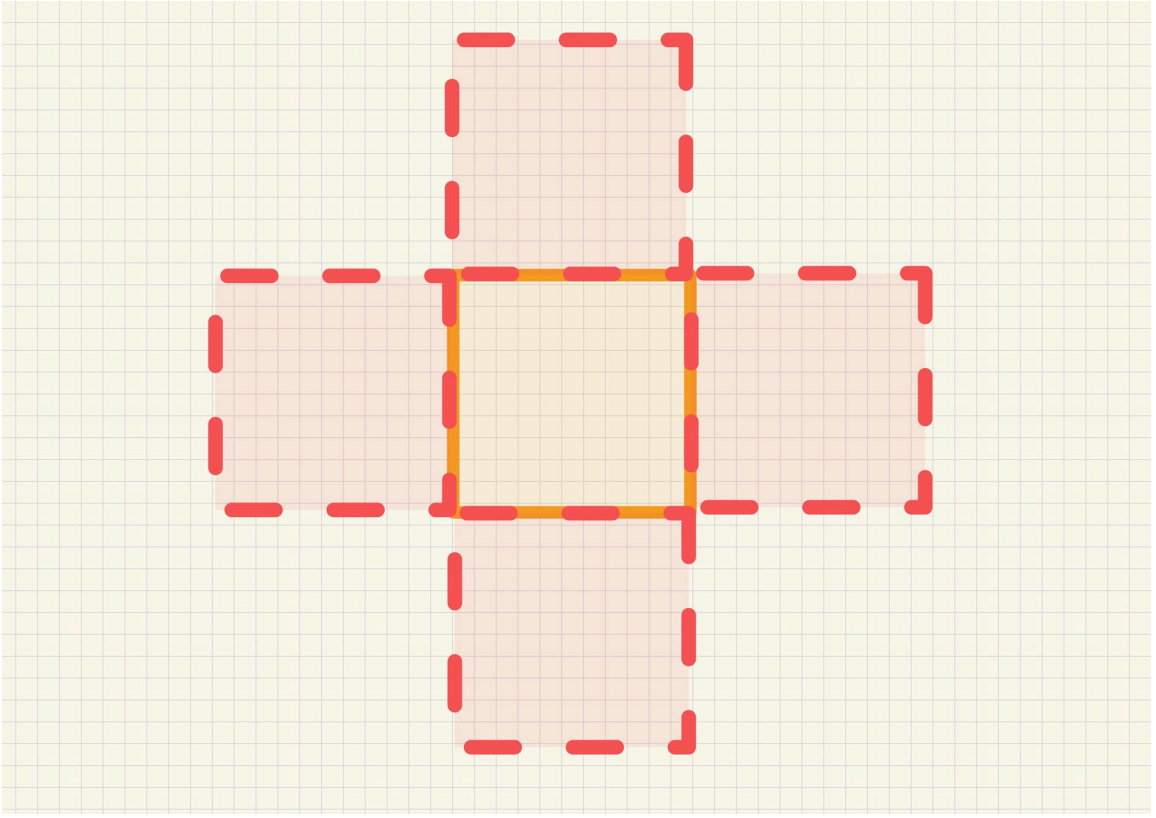
6. Anti - Defense
    - Similar to the defenses, after each turn anti-defenses health will drop.
    - Their main objective is to destroy the enemy defenses placed.
    - They will do damage each turn played.
    - The real main threat is the defenses if they decided to attack the Anti-Defenses directly.

# Details On Mechanics :

## Basic Movement

- King Movement
    - By default, the player can only one block in x and y axis.
    - Similar to the movement of king in the famous board game of chess.



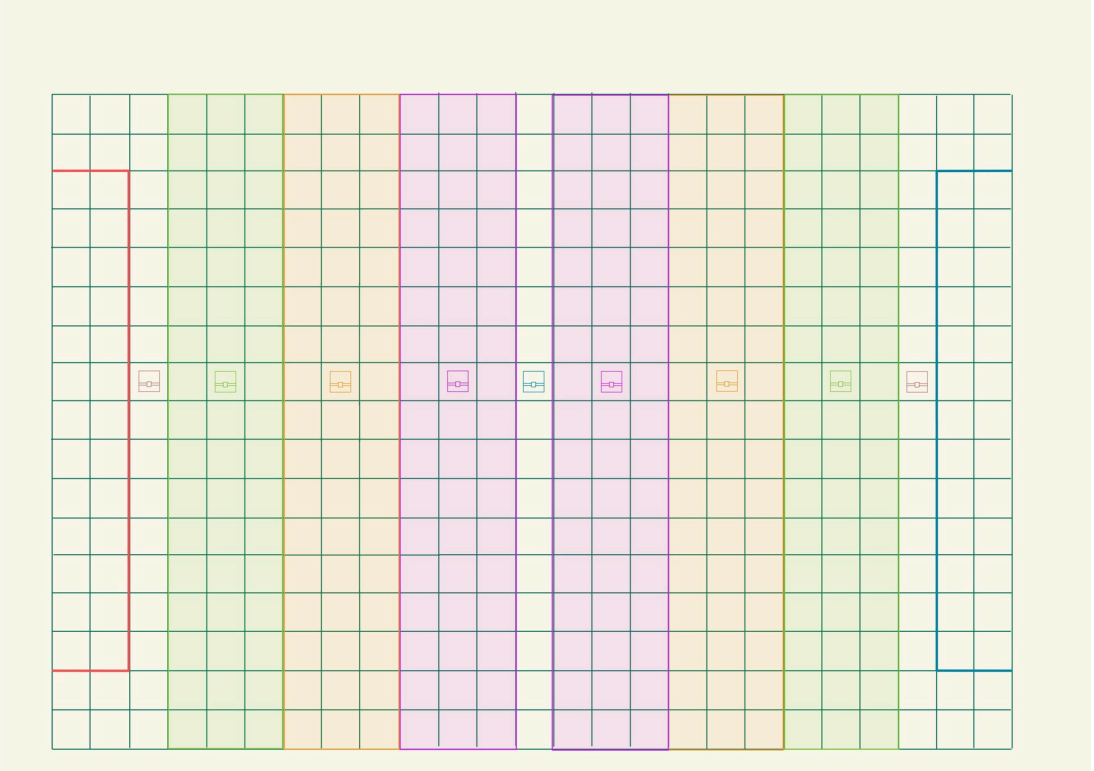


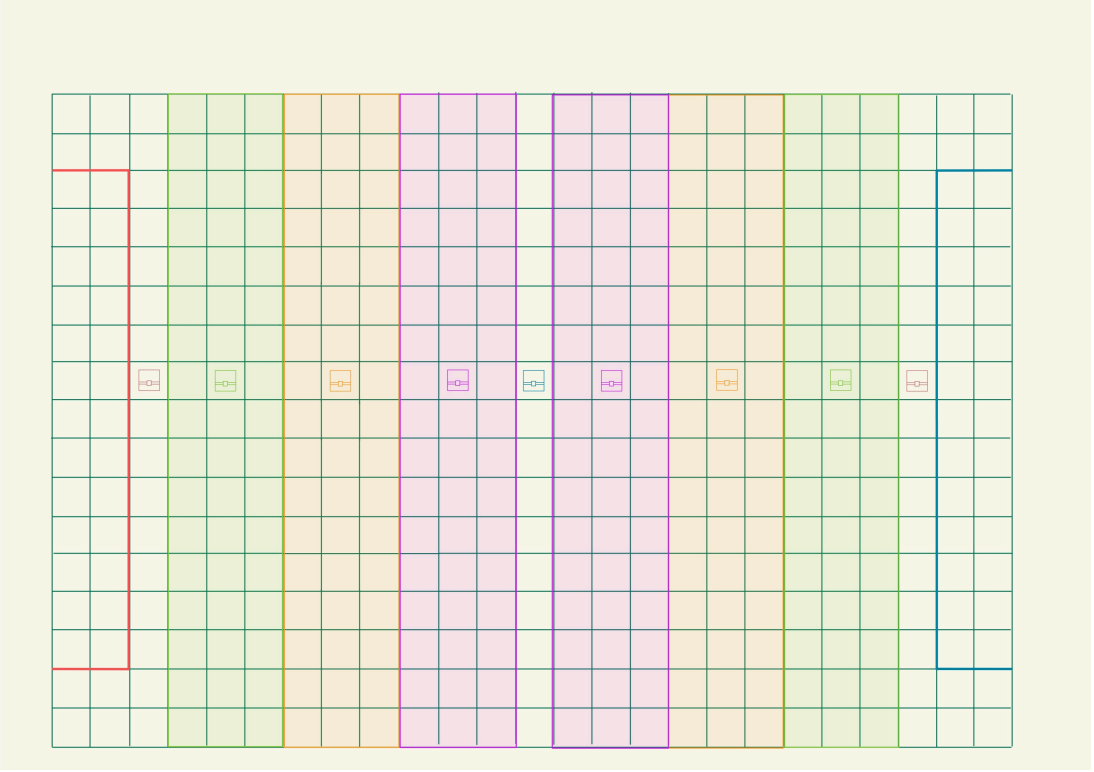
## Turn based play

- Once player 1 plays his turn it ends their turn.
- After player 1's turn player 2 plays their turn.

## Collectables/Reward System

- Chest
  - Chest Rarity
    - The Chests will have 5 rarities
      - Diamond - 10/10 power ups with 3 choices , coins.
      - Gem - 5/10 power ups with 2 choices, 8/10 coins, .
      - Gold - 3/10 power ups with 2 choices, 5/10 coins.
      - Iron - 0/10 power ups with 0 choices, 3/10 coins.
      - Wood - 1/10 power ups with 0 choices, 0/10 coins.
  - Chest Spawning
    - Each Chest rarity has its own zone assigned to it, as such





- Chest Rewards System
  - The player then gets cards presented to them from which the player will be given a choice between 0,2 or 3 depending on the chest rarity.

## Defenses

- Defenses will be attacking the opponent's base by default.
- They can be assigned to hit a anti - defense if required.
- Automated Archer Tower :
  - HP - 50
  - Attack Speed - 1 turns
  - Damage - 10
  - Grid Space - 2 × 2
- Cannon :
  - HP - 100
  - Attack Speed - 3 turns
  - Damage - 50
  - Grid Space - 3 × 3
- Bomb Tower :
  - HP - 150
  - Attack Speed - 5 turns
  - Damage - 100
  - Grid Space - 4 × 4

## Anti - Defenses

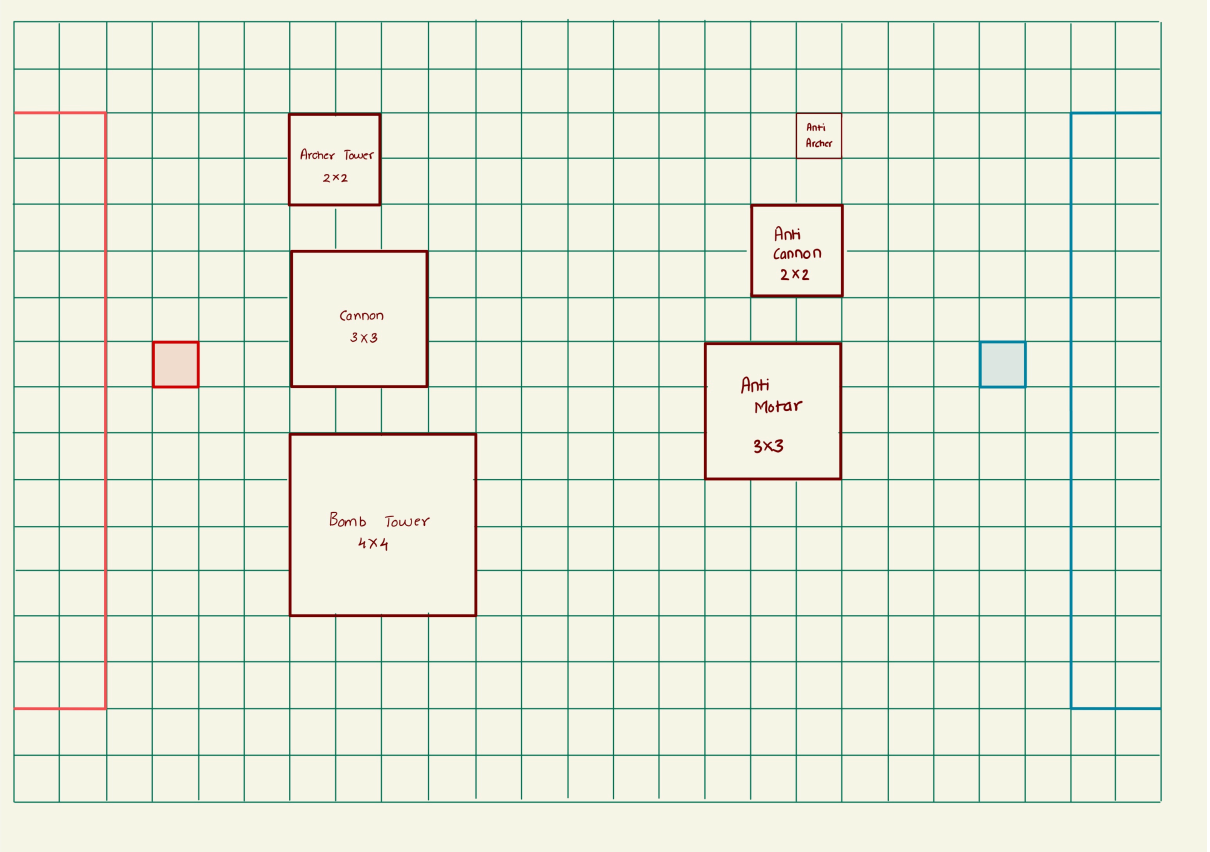
- Their main objective will be killing the defenses placed by the opponent.
- Anti - Defenses need to be assigned to a particular Defense when placing.
- If there are not any opponent defenses to target, the anti - defense will be placed with no action but later needs to be assigned to a defense to target.
- Archer destroyer :
  - HP - 20
  - Attack Speed - 2/1 times per turn

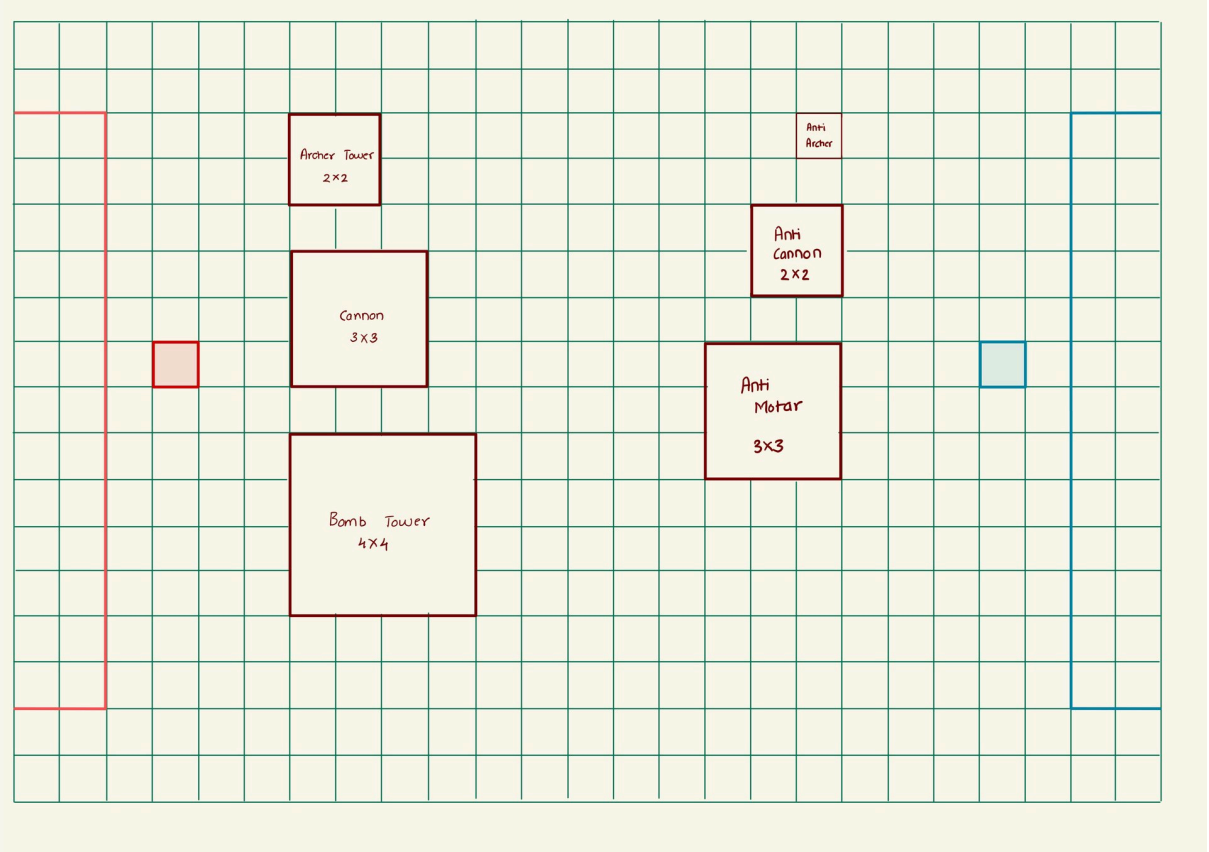
    - Damage - 10
- Cannon Destroyer:
    - HP - 50
    - Attack Speed - 1/1 times per turns
    - Damage - 25
- Anti - Bomb Tower :
    - HP - 100
    - Attack Speed - 1/2 turns
    - Damage - 50

## Resource Management

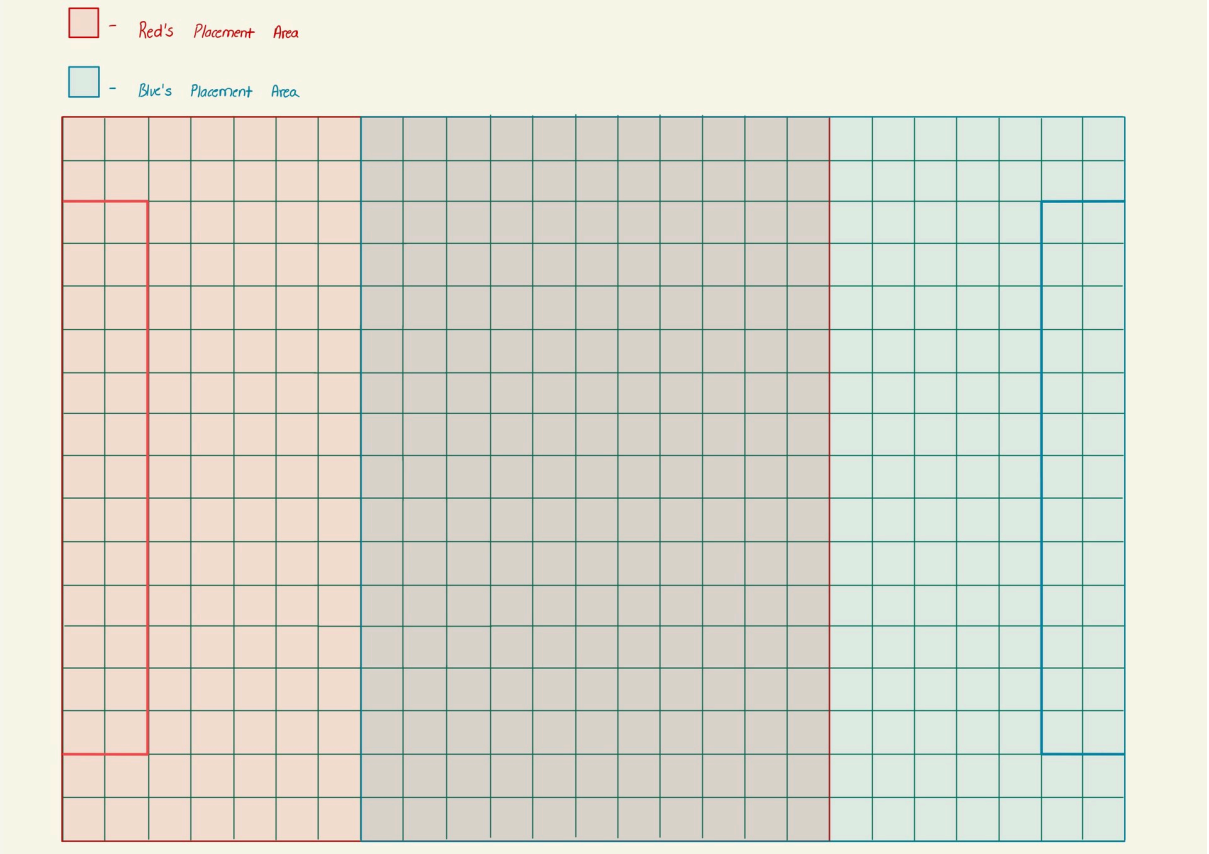
- Coin Generator
    - Level 1 : 0.5/3 coins per seconds
    - Level 2 : 0.75/3 coins per seconds
    - Level 3 : 1/3 coins per seconds
    - Level 4 : 1/2 coins per seconds
    - Level 5 : 1/1 coins per seconds
- Store
    - Defenses
        - Archer Tower : 100 coins
        - Cannon : 200 coins
        - Mortar : 400 coins
    - Anti - Defenses
        - Anti - Archer Tower : 70 coins
        - Anti - Cannon : 100 coins
        - Anti - Mortar : 200 coins

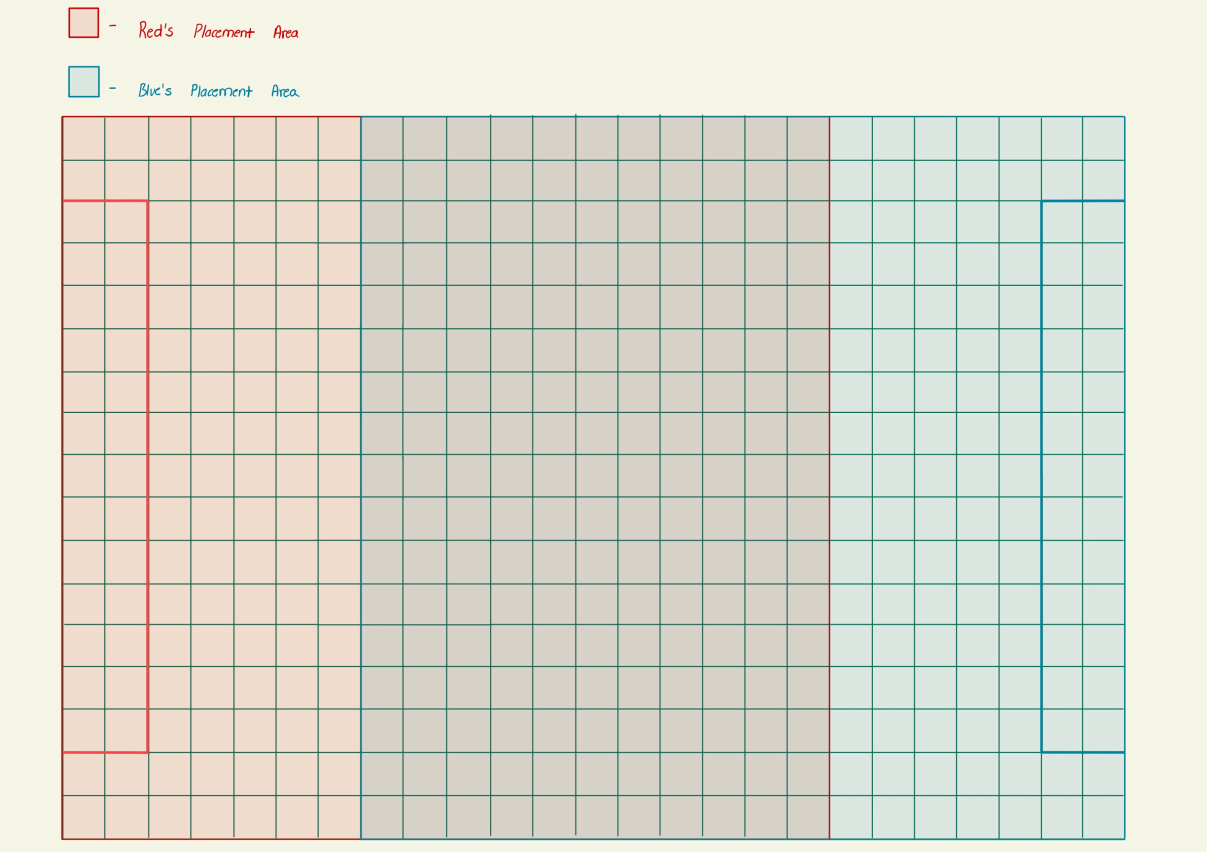
## Space required for each type of Defense and Anti - Defense on Grid Layout






## Defense and Anti - Defense Placement Zones for Red and Blue




- Red's Placement Area
- Blue's Placement Area.